

Periodic solutions for nonlinear differential equations in Banach spaces with generalized variable-exponent operators

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Motivation and the main problem

Periodic models in ordinary differential equations

Many systems (physical, biological, among others) are subject to recurrent effects, such as periodic forcing, cyclic loading, or oscillatory behaviours.

Usually, a simple way of modeling these phenomena is by second-order ordinary differential systems

$$u''(t) = f(t, u(t), u'(t)), \quad u : [0, T] \rightarrow \mathbb{R}^N.$$

For these models, we seek solutions satisfying

$$u(T) = u(0), \quad u'(T) = u'(0),$$

so that the system returns to the same state and velocity after one period.

Unlike IVP, the initial state is not prescribed: it must be chosen so that the trajectory closes after one period. Thus, periodic problems are difficult and have been a classical motivation for several methods of nonlinear analysis, including topological degree and continuation techniques.

From classical equations to nonlinear constitutive laws

The previous model assumes a linear response with respect to the velocity, since the principal term is u'' .

In nonlinear models, this response may instead be described by a constitutive law ϕ , leading to

$$\frac{d}{dt}\phi(u'(t)) = f(t, u(t), u'(t)).$$

Such problems arise, for instance, in non-Newtonian fluids and nonlinear elasticity, among others.

More generally, the constitutive response may also vary with time. For instance, a time-dependent coefficient may lead to

$$\frac{d}{dt}\left[a(t)\phi(u'(t))\right] = f(t, u(t), u'(t)),$$

Previous theory in finite dimensions

Both time-independent and time-dependent constitutive laws can be conveniently included in periodic problems of the form

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad u : [0, T] \rightarrow \mathbb{R}^N.$$

Problems driven by nonlinear operators of this type have been studied in several different settings, including recent works on generalized variable-exponent operators [8, 9, 13].

This naturally raises the question of extending this theory to infinite-dimensional state spaces.

From $\mathbb{R}^N$ to a Banach space

Such systems arise, for instance, in spatially distributed models, where the state at each time is itself a function:

$$u(t) = u(t, \cdot).$$

Therefore, the natural state space is no longer $\mathbb{R}^N$, but a Banach space X .

What is gained

The abstract equation can describe problems related to

- ▶ nonlinear PDEs;
- ▶ abstract differential equations;
- ▶ evolution problems.

What becomes more difficult

In infinite dimension:

- ▶ bounded sets need not be relatively compact;
- ▶ Some topological methods change drastically, or simply cease to exist.

The periodic problem studied in the thesis

Let X be a real separable reflexive Banach space, with dual X^* . We study

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), & t \in (0, T), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad (\text{P})$$

where

$$S : [0, T] \times X \rightarrow X^*, \quad f : [0, T] \times X \rightarrow X^*.$$

The same framework applies directly in a Hilbert space H , with $S, f : [0, T] \times H \rightarrow H$.

Definition of solution

A solution of (P) is a function $u \in C^1([0, T]; X)$ such that

$$S(\cdot, u'(\cdot)) \in W^{1,1}([0, T]; X^*)$$

and the differential equation holds a.e. on $(0, T)$.

Basic notation and periodic spaces

- Throughout this presentation,

$$\langle x^*, x \rangle$$

denotes the duality pairing between X^* and X .

- For a Banach space Y , we write

$$C_T([0, T]; Y) := \{u \in C([0, T]; Y) : u(0) = u(T)\},$$

and

$$C_0([0, T]; Y) := \{v \in C([0, T]; Y) : v(0) = v(T) = 0\}.$$

Structural assumptions on S

We assume that $S : [0, T] \times X \rightarrow X^*$ satisfies:

(S0) Continuity, normalization, and periodicity. The map S is continuous and, for every $t \in [0, T]$ and $x \in X$,

$$S(t, x) = 0 \iff x = 0, \quad S(0, x) = S(T, x).$$

(S1a) Local uniform strict monotonicity. For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0, \infty) \rightarrow [0, \infty), \quad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that, for every $t \in [0, T]$ and $\|x\|_X, \|y\|_X \leq M$,

$$\langle S(t, x) - S(t, y), x - y \rangle \geq \omega_M(\|x - y\|_X).$$

(S1b) Uniform coercivity. There exists a continuous non-decreasing function

$$\alpha : [0, \infty) \rightarrow [0, \infty), \quad \alpha(r) \rightarrow \infty \text{ as } r \rightarrow \infty,$$

such that, for every $t \in [0, T]$ and $x \in X$,

$$\langle S(t, x), x \rangle \geq \alpha(\|x\|_X) \|x\|_X.$$

(S2) Boundedness on bounded sets. There exists a continuous function $\beta : [0, T] \times [0, \infty) \rightarrow [0, \infty)$ such that, for every $t \in [0, T]$ and $x \in X$,

$$\|S(t, x)\|_{X^*} \leq \beta(t, \|x\|_X).$$

An example: the $p(t)$ -Laplacian-type operator

Let H be a real Hilbert space and let $\mathcal{R} : H \rightarrow H^*$ be the Riesz isomorphism.

Assume that $p \in C([0, T])$ satisfies

$$1 < p_- \leq p(t) \leq p_+ < \infty, \quad p(0) = p(T).$$

Variable-exponent operator

Define

$$S(t, 0) := 0, \quad S(t, x) := \|x\|_H^{p(t)-2} \mathcal{R}x, \quad x \neq 0.$$

Then

$$S : [0, T] \times H \rightarrow H^*$$

satisfies **(S0)**–**(S2)**.

In the Hilbert-space formulation, we may work directly with

$$S(t, x) = \|x\|_H^{p(t)-2} x, \quad S : [0, T] \times H \rightarrow H.$$

Invertibility and properties of the inverse

Fix $t \in [0, T]$ and set

$$S_t := S(t, \cdot) : X \rightarrow X^*.$$

Under **(S0)–(S2)**, S_t is continuous, strictly monotone, and coercive. Hence, by the Minty–Browder theorem, S_t is bijective.

We denote its inverse by

$$\Gamma(t, \cdot) := S_t^{-1} : X^* \rightarrow X.$$

Properties of Γ

- ▶ $\Gamma : [0, T] \times X^* \rightarrow X$ is jointly continuous and strictly monotone;
- ▶ $\|\Gamma(t, y)\|_X \rightarrow \infty$ as $\|y\|_{X^*} \rightarrow \infty$, uniformly in t ;
- ▶ if $g_n \rightarrow g$ in $C([0, T]; X^*)$, then

$$\Gamma(\cdot, g_n(\cdot)) \rightarrow \Gamma(\cdot, g(\cdot)) \quad \text{in } C([0, T]; X).$$

Nonlinear Green operator and generalized reduction

Compatibility and first integration

Let u solve the *non-homogeneous periodic problem*,

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = h(t), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad h \in L^1([0, T]; X^*). \quad (\text{NH})$$

Define

$$Qh := \int_0^T h(t) dt, \quad H(h)(t) := \int_0^t h(s) ds.$$

Integrating from 0 to T gives the following **necessary condition**

$$Qh = S(T, u'(T)) - S(0, u'(0)) = 0.$$

Integrating the differential equation from 0 to s , for $s \in [0, T]$, then applying $\Gamma(s, \cdot)$ and finally integrating from 0 to t , yields

$$u(t) = u(0) + \int_0^t \Gamma(s, c + H(h)(s)) ds, \quad c \in X^*$$

Position periodicity and the implicit equation

Evaluating the representation formula at $t = T$, we obtain

$$u(T) - u(0) = \int_0^T \Gamma(t, c + H(h)(t)) dt.$$

Therefore,

$$u(T) = u(0) \iff \int_0^T \Gamma(t, c + H(h)(t)) dt = 0.$$

To express this condition, for each $g \in C([0, T]; X^*)$, define

$$\Phi_g : X^* \rightarrow X, \quad \Phi_g(c) := \int_0^T \Gamma(t, c + g(t)) dt.$$

Hence, the periodicity condition is equivalent to requiring the integration constant $c \in X^*$ to solve

$$\Phi_{H(h)}(c) = 0.$$

Solvability of the implicit equation

Key lemma

For every $g \in C([0, T]; X^*)$, there exists a unique $\widehat{c}(g) \in X^*$ such that

$$\Phi_g(\widehat{c}(g)) = 0.$$

Moreover, $\widehat{c}: C([0, T]; X^*) \rightarrow X^*$ is continuous and maps bounded sets into bounded sets.

The nonlinear Green operator

From the previous construction, the integration constant is $c = \widehat{c}(H(h))$. This leads us to define

$$\mathcal{T}(h)(t) := \int_0^t \Gamma\left(s, \widehat{c}(H(h)) + H(h)(s)\right) ds.$$

Since $\widehat{c}(H(h))$ solves $\Phi_{H(h)}(c) = 0$, we have

$$\mathcal{T}(h)(0) = \mathcal{T}(h)(T) = 0.$$

Therefore $\mathcal{T} : L^1([0, T]; X^*) \longrightarrow C_0([0, T]; X)$.

Solvability and representation

The problem (NH) has a solution if and only if there exists $u \in C_T([0, T]; X)$ such that

$$\begin{cases} u(t) = u(0) + \mathcal{T}(h)(t), & t \in [0, T] \\ Qh = 0. \end{cases}$$

From the main problem to an abstract system

We now return to the nonlinear periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases}$$

Assume that f is Carathéodory and satisfies the local growth conditions needed to define the Nemytskii operator $N_f : C([0, T]; X) \rightarrow L^1([0, T]; X^*)$, $N_f(u)(t) := f(t, u(t))$.

Applying the previous representation with $h = N_f(u)$, we obtain

$$\begin{cases} u = u(0) + \mathcal{T}(N_f(u)), \\ QN_f(u) = 0, \end{cases}$$

The compactness defect

In infinite dimension, $u \mapsto u(0)$ is not a completely continuous operator. Hence, the constant component $u(0)$ must be separated.

The generalized Lyapunov–Schmidt decomposition

For every $u \in C_T([0, T]; X)$, we have the following unique decomposition,

$$u = u(0) + (u - u(0)), \quad u(0) \in X, \quad u - u(0) \in C_0([0, T]; X).$$

Since $X \cap C_0([0, T]; X) = \{0\}$ (using X represented as the constant functions),

$$C_T([0, T]; X) = X \oplus C_0([0, T]; X).$$

Set

$$a := u(0), \quad v := u - a.$$

Substituting $u = a + v$ into the abstract system and cancelling the constant component a , we obtain

$$\boxed{\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0, \end{cases} \quad a \in X, \quad v \in C_0([0, T]; X).} \quad (\text{LS})$$

Two existence strategies

The two methods developed in this thesis differ in how they treat the averaged constraint

$$QN_f(a + v) = 0,$$

which is a possibly infinite-dimensional problem.

First method: eliminate the constant component

If the constraint determines

$$a = \hat{A}_f(v),$$

then the system reduces to a single compact fixed-point equation for v .

Second method: approximate the coupled system

When the constraint cannot be solved explicitly for a , we solve finite-dimensional approximations of the full system and then pass to the limit to recover a solution of the original problem.

First existence method: mean-admissible nonlinearities

Mean-admissibility and the averaged constraint

Let Y, Z be reflexive Banach spaces and $f : [0, T] \times Y \rightarrow Z$. For $v \in C_0([0, T]; Y)$, define

$$F_v(a) := \int_0^T f(t, a + v(t)) dt.$$

Definition

The map f is *mean-admissible* if, for every v ,

$$F_v(a) = 0$$

has a unique solution $a = \hat{A}_f(v)$, where $\hat{A}_f : C_0([0, T]; Y) \rightarrow Y$ is continuous and maps bounded sets into bounded sets.

In the generalized Lyapunov–Schmidt system, $QN_f(a + v) = F_v(a)$. Hence,

$$QN_f(a + v) = 0 \iff a = \hat{A}_f(v).$$

A sufficient criterion for mean-admissibility

The implicit equation for Γ suggests imposing analogous monotonicity and coercivity assumptions on f .

Proposition

Let Y be a real separable reflexive Banach space and let $f : [0, T] \times Y \rightarrow Y^*$ be continuous. Assume, uniformly in $t \in [0, T]$, that:

- (i) There exists a non-decreasing function $\eta : [0, \infty) \rightarrow [0, \infty)$, with $\eta(r) > 0$ for $r > 0$, such that

$$\langle f(t, x) - f(t, y), x - y \rangle \geq \eta(\|x - y\|_Y).$$

- (ii) There exist $\Theta : [0, \infty) \rightarrow \mathbb{R}$ and a non-decreasing function $\gamma : [0, \infty) \rightarrow [0, \infty)$ such that

$$\langle f(t, x), x \rangle \geq \Theta(\|x\|_Y) \|x\|_Y, \quad \|f(t, x)\|_{Y^*} \leq \gamma(\|x\|_Y),$$

and

$$\Theta(r) \rightarrow \infty, \quad \Theta(r)r - M\gamma(r) \rightarrow \infty \quad \text{for every } M > 0.$$

Then f is *mean-admissible*.

The criterion is sufficient, but not necessary

Let $T = 2\pi$, let H be a real Hilbert space, and consider

$$f(t, x) = m(t)x + b(t), \quad m(t) := 1 + 2 \sin t.$$

Since $m(t) < 0$ for some t , the map $f(t, \cdot)$ is not pointwise monotone.

However, the averaged equation

$$\int_0^{2\pi} f(t, a + v(t)) dt = 0$$

has the unique solution

$$\hat{A}_f(v) = -\frac{1}{2\pi} \left(\int_0^{2\pi} m(t)v(t) dt + \int_0^{2\pi} b(t) dt \right).$$

Fixed-point reduction and the compactness issue

If f is mean-admissible, then the averaged constraint satisfies

$$QN_f(a + v) = 0 \iff a = \hat{A}_f(v).$$

Therefore, the generalized Lyapunov–Schmidt system reduces to

$$\boxed{v = \mathcal{T}\left(N_f\left(\hat{A}_f(v) + v\right)\right), \quad v \in C_0([0, T]; X).} \quad (\text{FP}_X)$$

To apply topological methods such as Leray–Schauder degree, the right-hand side must define a completely continuous operator.

However, for an arbitrary Banach space X ,

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; X)$$

need not be completely continuous.

Recovering compactness through a Gelfand triple

To recover compactness, we assume that X is part of a compact Gelfand triple

$$X \xhookrightarrow{i} H \cong H^* \xhookrightarrow{i^*} X^*,$$

where H is a real Hilbert space and $i : X \hookrightarrow H$ is dense and compact.

H -valued periodic solutions

Let $f : [0, T] \times H \rightarrow H$. An H -valued periodic solution is a function

$$u = a + v \in C_T([0, T]; H), \quad a \in H, \quad v \in C_0([0, T]; X) \cap C^1([0, T]; X),$$

satisfying

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases}$$

The compact fixed-point problem

The compact embedding $X \hookrightarrow H$ allows us to regard the Green operator as

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; H).$$

Compactness lemma

The operator $\mathcal{T}$ is continuous and maps bounded sets into relatively compact subsets of $C_0([0, T]; H)$.

Consequently, the reduced equation in the Gelfand setting is

$$\boxed{v = \mathcal{T}\left(i^* N_f\left(\widehat{A}_f(v) + v\right)\right), \quad v \in C_0([0, T]; H).} \quad (\text{FP}_H)$$

Its right-hand side is completely continuous, so Leray–Schauder degree can be applied.

A continuation theorem

Theorem

Let $\Omega \subset C_T([0, T]; H)$ be bounded and open, and let $f : [0, T] \times H \rightarrow H$ be a *mean-admissible* Carathéodory function. Assume that:

1. for every $\lambda \in (0, 1)$, the problem

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = \lambda i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases}$$

has no H -valued periodic solution on $\partial\Omega$.

2. there exists $c \in H$ such that the constant function $t \mapsto c$ belongs to Ω and

$$\int_0^T f(t, c) dt = 0;$$

Then (P) admits an H -valued periodic solution $u \in \overline{\Omega}$.

Proof idea I: constraint reduction

- ▶ We will start by supposing that there are no H -valued solutions in $\partial\Omega$ for $\lambda = 1$, since otherwise we would be finished with the proof.

- ▶ Define the constraint set

$$M := \left\{ u \in C_T([0, T]; H) : \int_0^T f(t, u(t)) dt = 0 \right\}.$$

which is nonempty since $u(t) \equiv c \in M$.

- ▶ Mean-admissibility gives a parametrization

$$R : C_0([0, T]; H) \rightarrow M, \quad R(v) := \widehat{A}_f(v) + v.$$

Since $v(0) = 0$, one has $R^{-1}(u) = u - u(0)$, so R is a homeomorphism.

- ▶ Transfer the domain through R :

$$\Omega_0 := R^{-1}(\Omega \cap M).$$

Then Ω_0 is bounded and open. Moreover,

$$\widehat{A}_f(0) = c \quad \implies \quad 0 \in \Omega_0.$$

Proof idea II: compact homotopy and degree

- For $\lambda \in [0, 1]$, define

$$\mathcal{G}(\lambda, v) := \mathcal{T}\left(\lambda i^* N_f(R(v))\right).$$

The compactness of $\mathcal{T}$ implies that $\mathcal{G}$ is completely continuous.

- If $v = \mathcal{G}(\lambda, v)$ with $v \in \partial\Omega_0$ and $\lambda \in (0, 1]$, then $u := R(v)$ solves the λ -problem and belongs to $\partial\Omega$, contradicting the boundary assumption.
- At $\lambda = 0$, $\mathcal{G}(0, v) = \mathcal{T}(0) = 0$. Since $0 \in \Omega_0$, there are no boundary fixed points, and homotopy invariance gives

$$d_{LS}[I - \mathcal{G}(1, \cdot), \Omega_0, 0] = d_{LS}[I, \Omega_0, 0] = 1.$$

- Since the degree is different from zero, there exists $v \in \Omega_0$ with $v = \mathcal{G}(1, v)$. Therefore

$$u := R(v) = \widehat{A}_f(v) + v$$

is an H -valued periodic solution with $u \in \Omega$.

Application: a double-phase periodic problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be open, bounded with Lipschitz boundary, and set

$$X = W_0^{1,\beta}(\mathcal{O}) \hookrightarrow H = L^2(\mathcal{O}) \hookrightarrow X^* = W^{-1,\beta'}(\mathcal{O}), \quad \beta > 2.$$

Let $p \in C([0, 2\pi])$ such that

$$1 < p_- \leq p(t) \leq p_+ \leq \beta, \quad p(0) = p(2\pi),$$

and define

$$m(t) := 1 + 2 \sin t, \quad b \in C([0, 2\pi]; H).$$

Consider, in X^* ,

$$\begin{cases} \partial_t \left[-\operatorname{div} \left(|\nabla u_t|^{p(t)-2} \nabla u_t + |\nabla u_t|^{\beta-2} \nabla u_t \right) \right] = i^* \left(m(t)u + b(t) \right), \\ u|_{\partial\mathcal{O}} = 0, \quad u(0) = u(2\pi), \quad u_t(0) = u_t(2\pi). \end{cases}$$

All the assumptions of the continuation theorem can be verified. Therefore, the problem admits an H -valued periodic solution.

Second existence method: Hartman–Galerkin

From monotonicity to the Galerkin strategy

The monotone structure of S' suggests approaching the problem through monotone-operator methods.

In infinite dimension, a natural strategy is to approximate each problem by finite-dimensional problems, solve the corresponding projected problems and verify that this sequence converges to a solution of the original problem. This is the Galerkin method.

finite-dimensional approximation $\longrightarrow$ solutions u_N $\longrightarrow$ uniform estimates $\longrightarrow$ limit solution

Variable-exponent functional setting

Let $p \in C([0, T])$ satisfy

$$1 < p_- \leq p(t) \leq p_+ < \infty, \quad p(0) = p(T),$$

and define

$$p'(t) := \frac{p(t)}{p(t) - 1}.$$

We work in the reflexive space

$$V := W_T^{1,p(\cdot)}([0, T]; X) = \left\{ u \in W^{1,p(\cdot)}([0, T]; X) : u(0) = u(T) \right\}.$$

For this second existence method, $S : [0, T] \times X \rightarrow X^*$ is assumed to satisfy the following stronger hypotheses.

(H0) Continuity, normalization, and periodicity. The map S is continuous and satisfies

$$S(t, x) = 0 \iff x = 0, \quad S(0, x) = S(T, x),$$

for every $t \in [0, T]$ and $x \in X$.

Stronger hypotheses on the principal operator

(H1) Local uniform strict monotonicity. For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0, \infty) \rightarrow [0, \infty), \quad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that

$$\langle S(t, x) - S(t, y), x - y \rangle \geq \omega_M(\|x - y\|_X)$$

for every $t \in [0, T]$ and $\|x\|_X, \|y\|_X \leq M$.

(H2) Variable-exponent coercivity. There exists $c_0 > 0$ such that

$$\langle S(t, x), x \rangle \geq c_0 \|x\|_X^{p(t)}$$

for every $t \in [0, T]$ and $x \in X$.

(H3) Variable-exponent growth. There exists $c_1 > 0$ such that

$$\|S(t, x)\|_{X^*} \leq c_1 \left(1 + \|x\|_X^{p(t)-1}\right)$$

for every $t \in [0, T]$ and $x \in X$.

Assumptions on the nonlinearity

Let $m, k : [0, T] \times X \rightarrow X^*$ be Carathéodory maps and set

$$f(t, x) := m(t, x) + k(t, x).$$

(F1) Local growth. For every $r > 0$, there exists $\eta_r \in L^{p'(\cdot)}([0, T])$ such that, for a.e. $t \in [0, T]$,

$$\|m(t, x)\|_{X^*} + \|k(t, x)\|_{X^*} \leq \eta_r(t) \quad \text{whenever } \|x\|_X \leq r.$$

(F2_m) Monotonicity. For a.e. t ,

$$\langle m(t, x) - m(t, y), x - y \rangle \geq 0.$$

(F2_{ws}) Weak-to-strong continuity. For a.e. t ,

$$x_n \rightharpoonup x \implies k(t, x_n) \rightarrow k(t, x) \quad \text{in } X^*.$$

(F3_e) Hartman condition. There exists $R > 0$ such that, for a.e. $t \in [0, T]$,

$$\langle f(t, x), x \rangle \geq 0 \quad \text{whenever } \|x\|_X \geq R.$$

Main Hartman-type theorem

Theorem

Assume that S satisfies **(H0)–(H3)**, and that

$$f(t, x) = m(t, x) + k(t, x)$$

satisfies **(F1)**, **(F2_m)**, **(F2_{ws})**, and **(F3_e)**.

Then (P) admits at least one periodic solution $u \in C^1([0, T]; X)$.

The finite-dimensional framework

Since X is separable, choose a linearly independent sequence

$$(e_j)_{j \geq 1} \subset X, \quad \overline{\text{span}\{e_j : j \geq 1\}}^X = X,$$

and set

$$X_N := \text{span}\{e_1, \dots, e_N\}.$$

Endow X_N with the auxiliary inner product $(\cdot, \cdot)_N$ for which $(e_1, \dots, e_N)$ is orthonormal, and define

$$R_N y := \sum_{j=1}^N \langle y, e_j \rangle e_j, \quad y \in X^*,$$

$$S_N(t, x) := R_N S(t, x), \quad x \in X_N.$$

Projected S operator

For every N and t , $S_N(t, \cdot) : X_N \rightarrow X_N$ is a homeomorphism. Moreover, $\Gamma_N(t, \cdot) := (S_N(t, \cdot))^{-1}$ verifies all the properties of Γ .

The regularized projected problems

To obtain a strict exterior inequality while recovering the original problem as $N \rightarrow \infty$, set

$$\varepsilon_N := \frac{1}{N}, \quad \widetilde{f}_N(t, x) := f(t, x) + \varepsilon_N S(t, x).$$

We want to find solutions u_N for the problem

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = R_N \widetilde{f}_N(t, u_N(t)), \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T). \end{cases} \quad (\mathbf{P}_N)$$

Hence, we will consider, for $\lambda \in [0, 1]$, the one-parameter family of problems,

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = \lambda R_N \widetilde{f}_N(t, u_N(t)), \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T). \end{cases} \quad (\mathbf{P}_{N,\lambda})$$

Finding the projected solutions by degree theory

Write

$$u_N = a_N + v_N, \quad a_N = u_N(0), \quad v_N = u_N - a_N,$$

hence $a_N \in X_N$, $v_N \in E_N := C_0([0, T]; X_N)$, where $\|v\|_{E_N} := \max_{t \in [0, T]} \|v(t)\|_X$.

The projected perturbed problem $(P_{N, \lambda})$ is equivalent to

$$\begin{cases} v_N = \mathcal{T}_N \left(\lambda R_N N_{\tilde{f}_N} (a_N + v_N) \right), \\ \mathcal{Q}_N(a_N + v_N) = 0, \end{cases}$$

where

$$\mathcal{Q}_N(u) := R_N \int_0^T \tilde{f}_N(t, u(t)) dt.$$

Define the associated fixed-point operator $\mathcal{K}_{N, \lambda} : X_N \times E_N \rightarrow X_N \times E_N$ as:

$$\mathcal{K}_{N, \lambda}(a, v) := \left(a - \mathcal{Q}_N(a + v), \mathcal{T}_N \left(\lambda R_N N_{\tilde{f}_N} (a + v) \right) \right).$$

This operator is completely continuous since X_N is finite-dimensional (standard argument using the properties of Γ_N and $\mathcal{Q}_N$).

Proof idea I: uniform Hartman bounds

- Let u_N solve $(P_{N,\lambda})$. Testing the equation with u_N gives

$$c_0 \int_0^T \|u'_N\|_X^{p(t)} dt \leq -\lambda \int_0^T \langle \tilde{f}_N(t, u_N), u_N \rangle dt.$$

- The Hartman condition and the strict regularization imply

$$\exists \tau_N \in [0, T] \quad \|u_N(\tau_N)\|_X < R.$$

Moreover, the integrand is nonpositive when $\|u_N(t)\|_X \geq R$, while **(F1)** controls it inside the ball, hence giving a uniform bound that only depends on R and **(F1)**.

- Consequently $\|u'_N\|_{L^{p(\cdot)}([0,T];X)} \leq C$, and by the FTC,

$$\|u_N(t)\|_X \leq \|u_N(\tau_N)\|_X + \int_{\tau_N}^t \|u'_N(s)\|_X ds.$$

Using $\|u_N(\tau_N)\|_X < R$ and Hölder's inequality, we obtain, for a constant independent of N and λ ,

$$\|u_N\|_{C([0,T];X)} \leq C \quad \implies \quad \|u_N\|_V \leq C'.$$

Proof idea II: degree and projected solutions

The uniform estimate for u_N over $C([0, T]; X)$ gives

$$\|a_N\|_X + \|v_N\|_{E_N} \leq C_1.$$

Choose $\rho > \max\{C_1, R\}$, and define the following open bounded set

$$\Omega_N := B_\rho^{X_N} \times B_\rho^{E_N} \subset X_N \times E_N,$$

where $B_\rho^{X_N} := \{a \in X_N : \|a\|_X < \rho\}$.

- ▶ No fixed point of $\mathcal{K}_{N,\lambda}$ lies on $\partial\Omega_N$ for $\lambda \in (0, 1]$.
- ▶ At $\lambda = 0$ the fixed point problem is $(a, v) = (a - \mathcal{Q}_N(a + v), 0)$; hence, the problem is equivalent to $\mathcal{Q}_N(a) = 0$. By an auxiliary homotopy and the exterior inequality, we get

$$d_{LS}[I - \mathcal{K}_{N,1}, \Omega_N, 0] = d_{LS}[I - \mathcal{K}_{N,0}, \Omega_N, 0] = d_B[\mathcal{Q}_N, B_\rho^{X_N}, 0] = 1.$$

- ▶ Hence $\mathcal{K}_{N,1}$ has a fixed point, and the regularized projected problem admits a solution u_N .

Proof idea III: passage to the limit

- ▶ Since $\|u_N\|_V \leq C$ uniformly, and V is reflexive, up to a subsequence,

$$u_N \rightharpoonup u \quad \text{in } V, \quad u_N(t) \rightharpoonup u(t) \quad \text{in } X.$$

- ▶ Weak-to-strong continuity and local growth give

$$k(\cdot, u_N) \rightarrow k(\cdot, u) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*), \quad \varepsilon_N S(\cdot, u_N) \rightarrow 0.$$

- ▶ Using the monotonicity of S and m , Minty's trick identifies the weak limit and yields

$$\int_0^T \langle S(t, u'), \varphi' \rangle dt + \int_0^T \langle f(t, u), \varphi \rangle dt = 0, \quad \forall \varphi \in V.$$

- ▶ Thus, u is a weak periodic solution. Setting $q := S(\cdot, u')$, we have

$$q \in W^{1, p'(\cdot)}([0, T]; X^*), \quad u'(t) = \Gamma(t, q(t)) \in C([0, T]; X),$$

and therefore $u \in C^1([0, T]; X)$. The periodicity of u follows from the same equation.

Two limiting cases

Purely monotone nonlinearity

Assume that S satisfies **(H0)–(H3)**, and that $f : [0, T] \times X \rightarrow X^*$ satisfies **(F1)** and is monotone in its second variable. If, for some $R > 0$,

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for a.e. } t \in [0, T] \text{ and } \|x\|_X = R,$$

then the periodic problem admits a solution. Any two solutions differ by a constant element of X ; if $f(t, \cdot)$ is strictly monotone, the solution is unique.

Purely weak-to-strong nonlinearity

Suppose that X is infinite-dimensional. If f satisfies **(F1)**, is sequentially weak-to-strong continuous, and

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for } \|x\|_X \geq R,$$

then every periodic solution is stationary. In particular, $u \equiv 0$ is a solution.

Radial Hilbert-space operators

Let X be a Hilbert space and assume that S satisfies **(H0)–(H3)** and has the radial form

$$S(t, 0) = 0, \quad S(t, x) = \beta(t, \|x\|_X) \mathcal{R}x, \quad \beta(t, r) > 0.$$

Proposition

Under the hypotheses of the main Hartman-type theorem, the periodic problem admits a solution u satisfying

$$\|u(t)\|_X \leq R, \quad t \in [0, T].$$

This partially extends the corresponding finite-dimensional radial Hartman bound of [8] to the Hilbert-space setting.

Application: a nonlocal Sobolev-type problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be defined as the previous application, $X = H_0^1(\mathcal{O})$, $X^* = H^{-1}(\mathcal{O})$, $A = -\Delta$ (in weak form), and

$$S(t, x) = \|x\|_X^{p(t)-2} Ax.$$

Let $b \in C([0, T]; H^{-1}(\mathcal{O}))$ be nonconstant and set

$$B := \|b\|_{C([0, T]; H^{-1}(\mathcal{O}))} < 1.$$

We study

$$\begin{cases} \partial_t \left[-\|\nabla u_t(t, \cdot)\|_2^{p(t)-2} \Delta_x u_t(t, x) \right] = -\frac{\Delta_x u(t, x)}{\sqrt{1 + \|\nabla u(t, \cdot)\|_2^2}} + b(t), \\ u|_{\partial\mathcal{O}}(t, \cdot) = 0, \quad u(0) = u(T), \quad u_t(0) = u_t(T). \end{cases}$$

Then there exists a unique nonstationary periodic solution satisfying

$$\|\nabla u(t, \cdot)\|_{L^2(\mathcal{O})} \leq \frac{B}{\sqrt{1 - B^2}}, \quad t \in [0, T].$$

Concluding remarks

Conclusions

- ▶ We introduced a broad class of monotone operators S , including p -Laplacian, $p(t)$ -Laplacian, and double-phase type operators.
- ▶ We derived a nonlinear Green operator for the associated nonhomogeneous periodic problem,
- ▶ For the nonlinear periodic problem, we found an equivalent formulation in the form of a generalized Lyapunov–Schmidt system.
- ▶ Two existence mechanisms were developed:

mean-admissibility

Hartman–Galerkin

- ▶ The results cover many types of problems, involving different nonlinearities and differential operators.

Open problems and future directions

Derivative-dependent nonlinearities

Can the existence methods developed here be extended to

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t))?$$

This would require new compactness arguments, a priori estimates for u' , and possibly additional structural conditions on f .

Alternative criteria for mean-admissibility

Are there other sufficient conditions for mean-admissibility beyond the pointwise monotonicity and coercivity assumptions considered here?

Further applications of the decomposition

Can the decomposition underlying the generalized Lyapunov–Schmidt system be adapted to other problems with a noncompact or infinite-dimensional constant component?

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Thank you

Questions

Periodic solutions for nonlinear differential equations in Banach spaces with generalized variable-exponent operators

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Appendix

Can the theory cover the classical wave equation?

Consider

$$\begin{cases} u_{tt}(t, x) = u_{xx}(t, x), & (t, x) \in [0, T] \times (0, 1), \\ u(t, 0) = u(t, 1) = 0, & t \in [0, T], \\ u(0, \cdot) = u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot). \end{cases}$$

If $H = L^2(0, 1)$, we may write formally

$$\frac{d}{dt}u_t = u_{xx}, \quad S(v) = v.$$

Thus $S : H \rightarrow H$ satisfies **(S0)**–**(S2)**.

However,

$$u \longmapsto u_{xx}$$

is not defined on all of $L^2(0, 1)$, since

$$D(\partial_{xx}) = H^2(0, 1) \cap H_0^1(0, 1).$$

Hence the L^2 case is not directly covered.

Can the theory cover the classical wave equation?

Let

$$H = H_0^1(0, 1), \quad (u, v)_H := \int_0^1 u_x v_x \, dx,$$

and define $B : H \rightarrow H$ by

$$(Bz, v)_H := \int_0^1 zv \, dx.$$

The weak formulation becomes

$$Bu_{tt} + u = 0, \quad \text{so } S(v) = Bv.$$

Take

$$\psi_n(x) = \frac{\sqrt{2}}{n\pi} \sin(n\pi x), \quad \|\psi_n\|_H = 1.$$

Then

$$(B\psi_n, \psi_n)_H = \frac{1}{(n\pi)^2} \rightarrow 0.$$

Hence B is not coercive in H , and the classical wave equation is not directly covered by the present framework.

The key lemma: existence and uniqueness

For $g \in C([0, T]; X^*)$, define

$$\Phi_g(c) := \int_0^T \Gamma(t, c + g(t)) dt, \quad \Psi_g := J_X \circ \Phi_g : X^* \rightarrow X^{**}.$$

Continuity of Γ makes Ψ_g continuous, hence hemicontinuous. Moreover, Ψ_g is strictly monotone, since for $c_1 \neq c_2$,

$$\langle \Psi_g(c_1) - \Psi_g(c_2), c_1 - c_2 \rangle = \int_0^T \langle c_1 - c_2, \Gamma(t, c_1 + g(t)) - \Gamma(t, c_2 + g(t)) \rangle dt > 0.$$

Set $z_c(t) := \Gamma(t, c + g(t))$. Coercivity of S gives

$$\langle \Psi_g(c), c \rangle \geq \int_0^T [\alpha(\|z_c(t)\|_X) - \|g\|_\infty] \|z_c(t)\|_X dt.$$

As $\|c\|_{X^*} \rightarrow \infty$, $\|z_c(t)\|_X \rightarrow \infty$ uniformly in t ; hence

$$\langle \Psi_g(c), c \rangle \rightarrow +\infty.$$

Since X^* is reflexive, the monotone Bolzano theorem [18, Theorem 4] yields a unique zero.

The key lemma: boundedness on bounded sets

Let $\Lambda \subset C([0, T]; X^*)$ be bounded and choose $M > 0$ such that

$$\|g\|_\infty \leq M \quad \text{for every } g \in \Lambda.$$

Write

$$c := \widehat{c}(g), \quad z(t) := \Gamma(t, c + g(t)).$$

Since $\Phi_g(c) = 0$,

$$\int_0^T \langle c + g(t), z(t) \rangle dt = \int_0^T \langle g(t), z(t) \rangle dt.$$

If $\|c\|_{X^*}$ were arbitrarily large, then $\|z(t)\|_X$ would be uniformly large, so that

$$\alpha(\|z(t)\|_X) \geq 2M.$$

Consequently,

$$2M \int_0^T \|z(t)\|_X dt \leq \int_0^T \langle c + g(t), z(t) \rangle dt \leq M \int_0^T \|z(t)\|_X dt,$$

a contradiction. Therefore $\widehat{c}: C([0, T]; X^*) \rightarrow X^*$ maps bounded sets into bounded sets.

The key lemma: continuity of $\widehat{c}$

Let $g_n \rightarrow g$ in $C([0, T]; X^*)$, and set

$$c_n := \widehat{c}(g_n), \quad c := \widehat{c}(g), \quad \Delta_n(t) := \Gamma(t, c_n + g_n(t)) - \Gamma(t, c + g(t)).$$

The preceding result makes (c_n) bounded; hence both values of Γ lie in a fixed ball $B_{M_B}^X$. Using

$$\Phi_{g_n}(c_n) = \Phi_g(c) = 0$$

and (S1), we obtain

$$\int_0^T \omega_{M_B}(\|\Delta_n(t)\|_X) dt \leq 2M_B T \|g_n - g\|_\infty \rightarrow 0.$$

Thus every subsequence has a further subsequence (n_k) and some $t_0 \in [0, T]$ such that $\Delta_{n_k}(t_0) \rightarrow 0$ in X .

Since $S(t_0, \Gamma(t_0, c_{n_k} + g_{n_k}(t_0))) = c_{n_k} + g_{n_k}(t_0)$, the continuity of $S(t_0, \cdot)$ and $g_{n_k}(t_0) \rightarrow g(t_0)$ imply

$$c_{n_k} \rightarrow c \quad \text{in } X^*.$$

Since every subsequence of c_n has a further subsequence that converges to c , this means that the entire sequence converges, hence $\widehat{c}$ is continuous.

Why does the Green operator become compact in H ?

Assume the compact Gelfand triple

$$X \hookrightarrow H \cong H^* \hookrightarrow X^*.$$

For a bounded sequence $(h_n) \subset L^1([0, T]; X^*)$, write

$$v_n := \mathcal{T}(h_n).$$

The estimates for Γ give:

- ▶ (v_n) is equicontinuous in H ;
- ▶ for each t , $(v_n(t))$ is bounded in X , hence relatively compact in H .

Therefore, Arzelà–Ascoli yields

$$\mathcal{T} : L^1([0, T]; X^*) \longrightarrow C_0([0, T]; H) \text{ completely continuous.}$$

Why is the projected problem regularized?

In the Galerkin problems, set

$$\widetilde{f}_N(t, x) = f(t, x) + \varepsilon_N S(t, x), \quad \varepsilon_N = \frac{1}{N}.$$

Then, for $\|x\|_X \geq R$,

$$\langle \widetilde{f}_N(t, x), x \rangle = \langle f(t, x), x \rangle + \varepsilon_N \langle S(t, x), x \rangle > 0.$$

Thus, the regularization:

- ▶ gives the strict exterior inequality needed for the degree;
- ▶ excludes solutions on the boundary of the Galerkin domain;
- ▶ disappears in the limit, since

$$\varepsilon_N S(\cdot, u_N) \rightarrow 0.$$

Minty's trick I: where does the inequality come from?

Let $u_N \rightharpoonup u$ in V . For any $w \in V$, choose $w_N \in V_N$ such that $w_N \rightarrow w$ strongly in V .

Testing the projected equation with $u_N - w_N \in V_N$ gives

$$\begin{aligned} 0 = & \int_0^T \langle S(t, u'_N), u'_N - w'_N \rangle dt + \int_0^T \langle m(t, u_N), u_N - w_N \rangle dt \\ & + \int_0^T \langle k(t, u_N), u_N - w_N \rangle dt + \varepsilon_N \int_0^T \langle S(t, u_N), u_N - w_N \rangle dt. \end{aligned}$$

By monotonicity of $S(t, \cdot)$ and $m(t, \cdot)$,

$$\begin{aligned} 0 \geq & \int_0^T \langle S(t, w'_N), u'_N - w'_N \rangle dt + \int_0^T \langle m(t, w_N), u_N - w_N \rangle dt \\ & + \int_0^T \langle k(t, u_N), u_N - w_N \rangle dt + \varepsilon_N \int_0^T \langle S(t, u_N), u_N - w_N \rangle dt. \end{aligned}$$

Using $w_N \rightarrow w$ in V , the weak-to-strong convergence of k , and the vanishing of the regularization, we obtain

$$\boxed{0 \geq \int_0^T \langle S(t, w'), u' - w' \rangle dt + \int_0^T \langle m(t, w) + k(t, u), u - w \rangle dt} \quad (w \in V).$$

Minty's trick II: recovering the limit equation

Starting from

$$0 \geq \int_0^T \langle S(t, w'), u' - w' \rangle dt + \int_0^T \langle m(t, w) + k(t, u), u - w \rangle dt, \quad w \in V,$$

take $w = u - \lambda v$, with $v \in V$ and $\lambda > 0$. Dividing by λ gives

$$0 \geq \int_0^T \langle S(t, u' - \lambda v'), v' \rangle dt + \int_0^T \langle m(t, u - \lambda v) + k(t, u), v \rangle dt.$$

Letting $\lambda \rightarrow 0^+$, continuity yields

$$\int_0^T \langle S(t, u'), v' \rangle dt + \int_0^T \langle m(t, u) + k(t, u), v \rangle dt \leq 0.$$

Replacing v by $-v$ gives the reverse inequality. Therefore

$$\boxed{\int_0^T \langle S(t, u'), v' \rangle dt + \int_0^T \langle f(t, u), v \rangle dt = 0 \quad \text{for every } v \in V.}$$

Thus u is a weak solution of the limit problem.

Why is the weak solution actually C^1 ?

After the passage to the limit, define

$$q(t) := S(t, u'(t)).$$

The equation gives

$$q' \in L^{p'(\cdot)}([0, T]; X^*), \quad q \in W^{1, p'(\cdot)}([0, T]; X^*).$$

Since the time interval is one-dimensional,

$$q \in C([0, T]; X^*).$$

Using the continuous inverse of $S(t, \cdot)$,

$$u'(t) = \Gamma(t, q(t)) \in C([0, T]; X).$$

Hence

$$u \in C^1([0, T]; X).$$

Can the mean-admissible method include u' ?

Consider

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t)).$$

The reduced equation would involve

$$v = \mathcal{T}\left(N_f\left(\widehat{A}_f(v) + v\right)\right),$$

and must be studied in a C^1 -type space.

However, the present argument only proves compactness of

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; H).$$

Indeed,

$$(\mathcal{T}h)'(t) = \Gamma\left(t, \widehat{c}(H(h)) + H(h)(t)\right),$$

and the current estimates give uniform boundedness, but not the equicontinuity or relative compactness of this derivative family.

The first method therefore does not extend directly.

A possible Galerkin extension with dependence on u'

A possible route is to work in a Hilbert space H and consider

$$\frac{d}{dt}S(t, u') = f(t, u, u').$$

Inspired by classical Hilbert-space results, one could impose the joint monotonicity condition

$$\left(S(t, y) - S(t, z), y - z \right)_H + \left(f(t, x, y) - f(t, w, z), x - w \right)_H \geq 0$$

for every $x, w, y, z \in H$.

A natural Hartman condition would be

$$\left(S(t, y), y \right)_H + \left(f(t, x, y), x \right)_H \geq 0, \quad \|x\|_H \geq R.$$

When $S = I$, these terms become

$$\|y - z\|_H^2, \quad \|y\|_H^2 + \left(f(t, x, y), x \right)_H,$$

as in the classical second-order Hilbert-space setting shown in [17].

These conditions suggest a Galerkin–Minty extension, but a complete proof would require new growth estimates and a careful limit argument.

Why is the purely monotone case stronger?

Assume that $f(t, \cdot)$ is monotone and that

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for } \|x\|_X = R.$$

For $\|x\|_X = r > R$, set

$$y := \frac{R}{r}x.$$

Then $\|y\|_X = R$, and monotonicity gives

$$0 \leq \langle f(t, x) - f(t, y), x - y \rangle = \left(1 - \frac{R}{r}\right) \langle f(t, x) - f(t, y), x \rangle.$$

Hence

$$\langle f(t, x), x \rangle \geq \frac{r}{R} \langle f(t, y), y \rangle \geq 0.$$

Thus the Hartman condition holds on the whole exterior $\|x\|_X \geq R$, so the main theorem applies.

If u, w are two solutions, testing the difference with $u - w$ gives

$$0 = \int_0^T \langle S(t, u') - S(t, w'), u' - w' \rangle dt + \int_0^T \langle f(t, u) - f(t, w), u - w \rangle dt.$$

Both terms are nonnegative, hence $u' = w'$, so $u - w$ is constant. Strict monotonicity of f gives $u = w$.

Why is the purely weak-to-strong case degenerate?

Assume that X is infinite-dimensional and that $f(t, \cdot)$ is weak-to-strong continuous.

Choose

$$e_j \rightharpoonup 0, \quad \|e_j\|_X = 1.$$

For fixed $x \in X$, take $L > R + \|x\|_X$. Then $\|x + Le_j\|_X > R$ so the Hartman condition gives

$$\langle f(t, x + Le_j), x + Le_j \rangle \geq 0.$$

Since

$$x + Le_j \rightharpoonup x, \quad f(t, x + Le_j) \rightarrow f(t, x) \quad \text{in } X^*,$$

passing to the limit yields

$$\boxed{\langle f(t, x), x \rangle \geq 0 \quad \text{for every } x \in X.}$$

Taking $x = \pm \varepsilon v$ and $\varepsilon \rightarrow 0$ gives $f(t, 0) = 0$. Finally, testing the equation with u ,

$$0 = \int_0^T \langle S(t, u'), u' \rangle dt + \int_0^T \langle f(t, u), u \rangle dt.$$

Both terms are nonnegative, hence $u' = 0$, and hence every solution is constant with respect to $t \in [0, T]$.

Radial case: proof idea

Set

$$\varphi_N(t) = \frac{1}{2} \|u_N(t)\|_X^2.$$

If σ is a maximum with $\|u_N(\sigma)\|_X > R$, then

$$\varphi'_N(\sigma) = 0, \quad (S_N(\sigma, u'_N(\sigma)), u_N(\sigma))_X = 0.$$

Hartman's condition implies, just after σ ,

$$(S_N(t, u'_N), u_N)_X > 0.$$

But radiality gives

$$(S_N(t, u'_N), u_N)_X = \beta(t, \|u'_N\|_X) \varphi'_N(t),$$

hence $\varphi'_N(t) > 0$, a contradiction.

Therefore

$$\|u_N(t)\|_X \leq R \implies \|u(t)\|_X \leq R.$$

Double-phase periodic problem

$$\begin{aligned} \frac{\partial}{\partial t} \left[-\operatorname{div} (|\nabla u_t|^{p(t)-2} \nabla u_t + |\nabla u_t|^{\beta-2} \nabla u_t) \right] &= f(t, u) \quad \text{in } (0, T) \times \Omega, \\ u(t, \cdot) &= 0 \quad \text{on } \partial\Omega, \\ u(0, \cdot) &= u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot). \end{aligned}$$

Set

$$X = W_0^{1,\beta}(\Omega), \quad 1 < p_- \leq p(t) \leq p_+ \leq \beta, \quad \beta > 2.$$

The spatial operator is understood weakly as $S(t, \cdot) : X \rightarrow X^*$, where

$$\langle S(t, v), \phi \rangle = \int_{\Omega} (|\nabla v|^{p(t)-2} \nabla v + |\nabla v|^{\beta-2} \nabla v) \cdot \nabla \phi \, dx, \quad \phi \in X.$$

Thus the PDE has the abstract form

$$\frac{d}{dt} S(t, u_t(t)) = f(t, u(t)) \quad \text{in } X^*.$$

Double-phase operator: verification of (S0) – (S2)

The continuity of $p(\cdot)$, $p(0) = p(T)$, and

$$S(t, 0) = 0, \quad S(t, v) = 0 \implies v = 0,$$

give (S0).

Since both phases are monotone,

$$\langle S(t, v) - S(t, w), v - w \rangle \geq C_\beta \|v - w\|_X^\beta,$$

while

$$\langle S(t, v), v \rangle \geq \|v\|_X^\beta.$$

Hence (S1) holds.

Moreover,

$$\|S(t, v)\|_{X^*} \leq C(1 + \|v\|_X^{\beta-1}),$$

uniformly in t , so (S2) follows.

A nonlocal $p(t)$ -Laplacian-type problem

$$\begin{aligned}\frac{\partial}{\partial t} \left[\|u_t(t)\|_{H_0^1(\Omega)}^{p(t)-2} (-\Delta u_t) \right] &= f(t, u) \quad \text{in } (0, T) \times \Omega, \\ u(t, \cdot) &= 0 \quad \text{on } \partial\Omega, \\ u(0, \cdot) &= u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot).\end{aligned}$$

Set

$$X = H_0^1(\Omega), \quad \langle Ax, \phi \rangle = \int_{\Omega} \nabla x \cdot \nabla \phi \, dx,$$

so $A : X \rightarrow X^*$ is the Riesz map associated with $-\Delta$.

Define $S(t, x) = \|x\|_X^{p(t)-2} Ax$, which is equivalent to

$$\langle S(t, x), \phi \rangle = \|x\|_X^{p(t)-2} \int_{\Omega} \nabla x \cdot \nabla \phi \, dx.$$

Therefore

$$\frac{d}{dt} S(t, u_t(t)) = f(t, u(t)) \quad \text{in } X^*.$$

$p(t)$ -Laplacian-type operator: verification of (H0) – (H3)

Continuity, $S(t, 0) = 0$, and $p(0) = p(T)$ give (H0).

On bounded sets,

$$\langle S(t, x) - S(t, y), x - y \rangle \geq c_M \min \left\{ \|x - y\|_X^2, \|x - y\|_X^{p_+} \right\},$$

hence (H1).

Moreover,

$$\langle S(t, x), x \rangle = \|x\|_X^{p(t)},$$

which gives (H2).

Finally,

$$\|S(t, x)\|_{X^*} = \|x\|_X^{p(t)-1},$$

hence (H3).

Pseudomonotone nonlinearities

A map $F : X \rightarrow X^*$ is **pseudomonotone** if

$$x_n \rightharpoonup x, \quad \limsup_{n \rightarrow \infty} \langle F(x_n), x_n - x \rangle \leq 0$$

imply, for every $y \in X$,

$$\langle F(x), x - y \rangle \leq \liminf_{n \rightarrow \infty} \langle F(x_n), x_n - y \rangle.$$

Relation with our assumptions

If

$$f = m + k,$$

where m is monotone and hemicontinuous and k is sequentially weak-to-strong continuous, then f is pseudomonotone.

Possible extension of the Galerkin approach

The decomposition

$$f = m + k$$

is mainly used to identify the nonlinear term when passing to the Galerkin limit.

Possible extension

One may instead assume directly that $f(t, \cdot) : X \rightarrow X^*$ is pseudomonotone for a.e. t .

Under suitable growth assumptions, one expects the corresponding Nemytskii operator on the periodic Sobolev space to be pseudomonotone, allowing the same weak-limit passage.

Thus the Hartman–Galerkin strategy should extend to

pseudomonotone f + Hartman-type a priori bounds.